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Department of CSE

B.Tech. CSE Trimester -5

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Section: 3

Cloud Infrastructure and Services
25CS2206E

SKILL RECORD

2026

Skill -1

AIM: To create an IAM user with programmatic access only, download the access key and secret key, and set up a billing alert for the account so that an email is received when the Free Tier limit is about to be crossed.

Description:

AWS Identity and Access Management (IAM) is a web service that is used to securely control access to AWS resources. It is used to create and manage AWS users and user groups, and to apply permissions that allow or deny their access to AWS resources. An IAM user is an identity that is created within an AWS account. Programmatic access allows a user to call AWS services through the AWS CLI, the AWS SDKs or direct API calls using an access key, without signing in to the AWS Management Console.

An access key consists of two parts, an Access Key ID and a Secret Access Key, which together act like a user name and password for the programmatic calls. The secret access key can be viewed or downloaded only once at the time of creation, so it has to be stored safely. While creating an IAM user, the permissions can be attached directly to the user, copied from an existing user, or the user can be added to a user group. In this skill, the AdministratorAccess AWS managed policy is attached to the user.

Since the account is a free tier account, unexpected charges have to be avoided. AWS Budgets in the Billing and Cost Management console is used to set up a billing alert. A zero spend budget sends an email notification to the given email recipients whenever the spending exceeds $0.01, so that the account owner is alerted before the free tier limit is crossed and any charges are incurred.

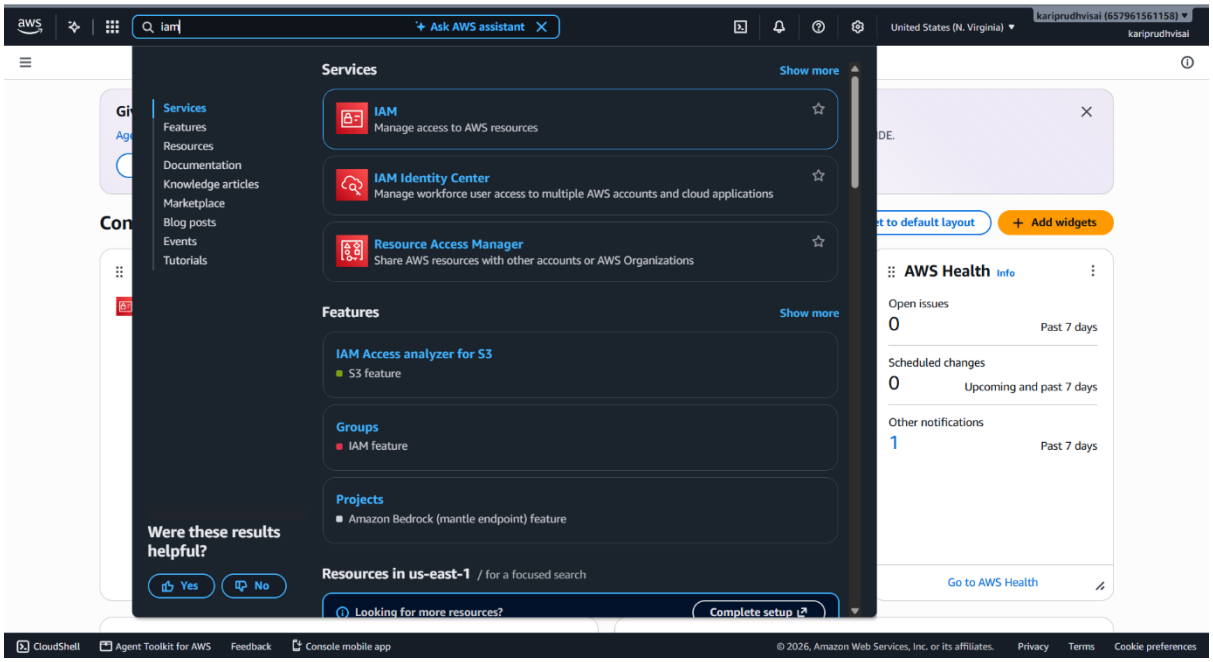
Steps:

1. Sign in to the AWS Management Console and open the IAM (Identity and Access Management) service.
2. In the IAM console, go to Access Management and choose IAM users, then click Create user.
3. In the Specify user details step, enter the user name as Shiva and keep the option Provide user access to the AWS Management Console unchecked, so that the user is created with programmatic access only. Then click Next.
4. In the Set permissions step, select Attach policies directly, search for the AdministratorAccess policy and select it. Then click Next.
5. In the Review and create step, verify the user details such as the user name, console password type as None and the attached AdministratorAccess policy, then click Create user.

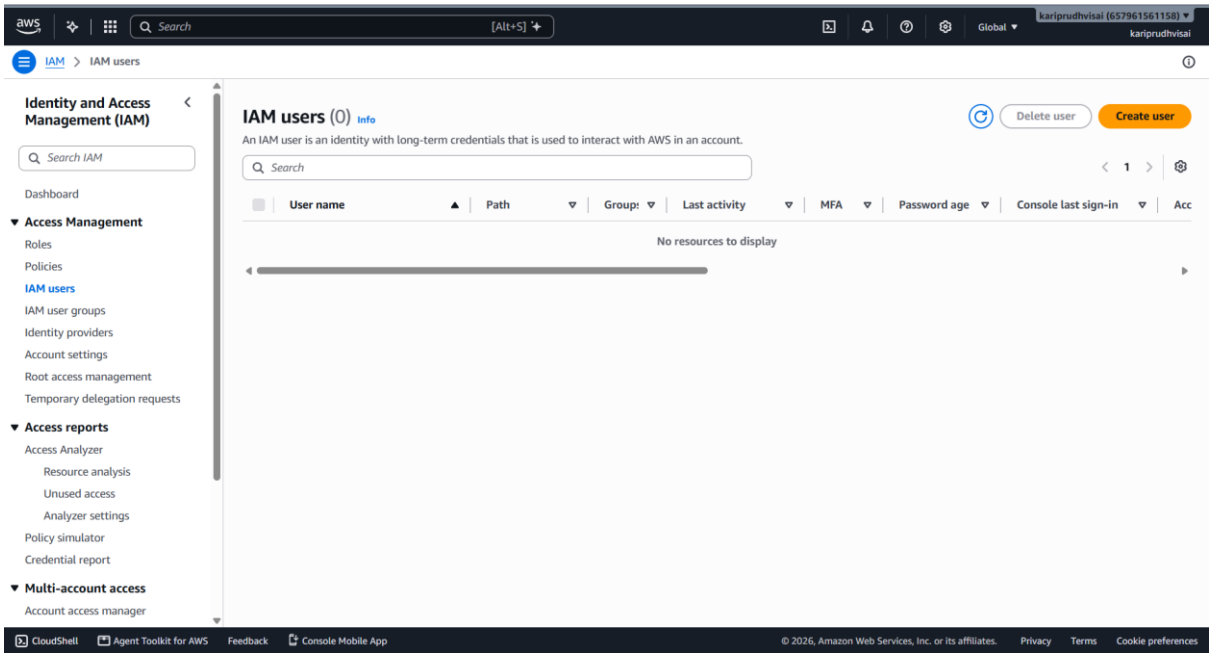
6. To test the created user, console access was enabled from the Security credentials tab and the auto generated console password was downloaded as a CSV file.
7. Sign in to the AWS console as the IAM user using the account ID 657961561158 and the IAM user name on the sign-in page.
8. After signing in, the Console Home page opened for the IAM user, which confirmed that the user was created correctly.
9. Open the IAM user Shiva, go to the Security credentials tab and click Create access key.
10. In the Access key best practices and alternatives step, select the Command Line Interface (CLI) use case and continue.
11. In the Retrieve access keys step, the Access Key ID and the Secret Access Key were displayed. The credentials were downloaded as a CSV file with the Download .csv file button, because the secret access key cannot be viewed again later.
12. The Security credentials page of the user K.prudhvisai@2008 showed Access keys (1) with the status Active.
13. A second IAM user New was created in the same way, with console access kept disabled, so that the user has programmatic access only.
14. For the user New, click Create access key, select the Command Line Interface (CLI) use case and proceed to retrieve the access keys.
15. The Access Key ID and Secret Access Key of the user New were displayed only once and were downloaded as a CSV file.
16. The Security credentials page of the user New confirmed that the access key was created with the status Active.
17. The IAM users page listed both the users New and Prudhvisai@2008 with their details.
18. To set up the billing alert, search for billing in the AWS console search bar and open Billing and Cost Management.
19. In the Billing console, go to Budgets and click Create budget. The Use a template (simplified) option with the Zero spend budget template was reviewed, and the Customize (advanced) option with the Cost budget was also reviewed.
20. Finally, the Zero spend budget template was selected. The budget name was entered as My Zero-Spend Budget and the email recipient was given as karkalashivareddy@gmail.com. With all AWS services in scope, an email notification is sent when any spend above $0.01 is incurred. Then click Create budget to complete the setup.

Result: Thus, an IAM user with programmatic access only was created successfully, the access key and secret access key were generated and downloaded, and a billing alert was set up using a zero spend budget so that an email is received when the spending is about to cross the free tier limit

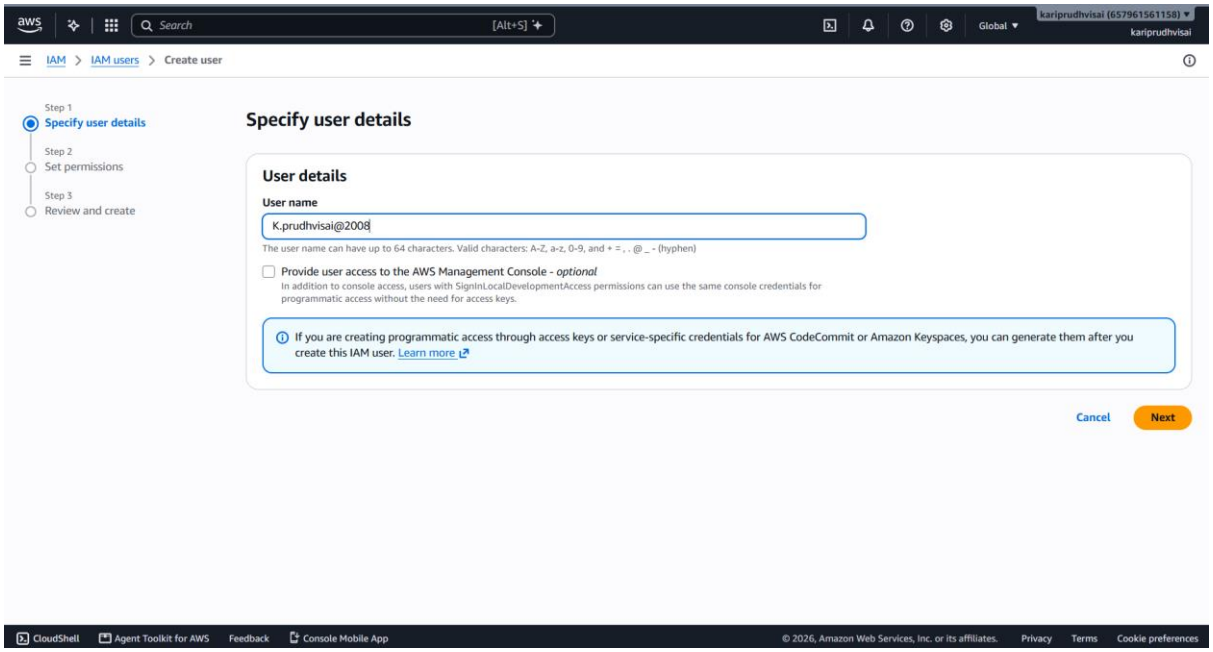
Output:



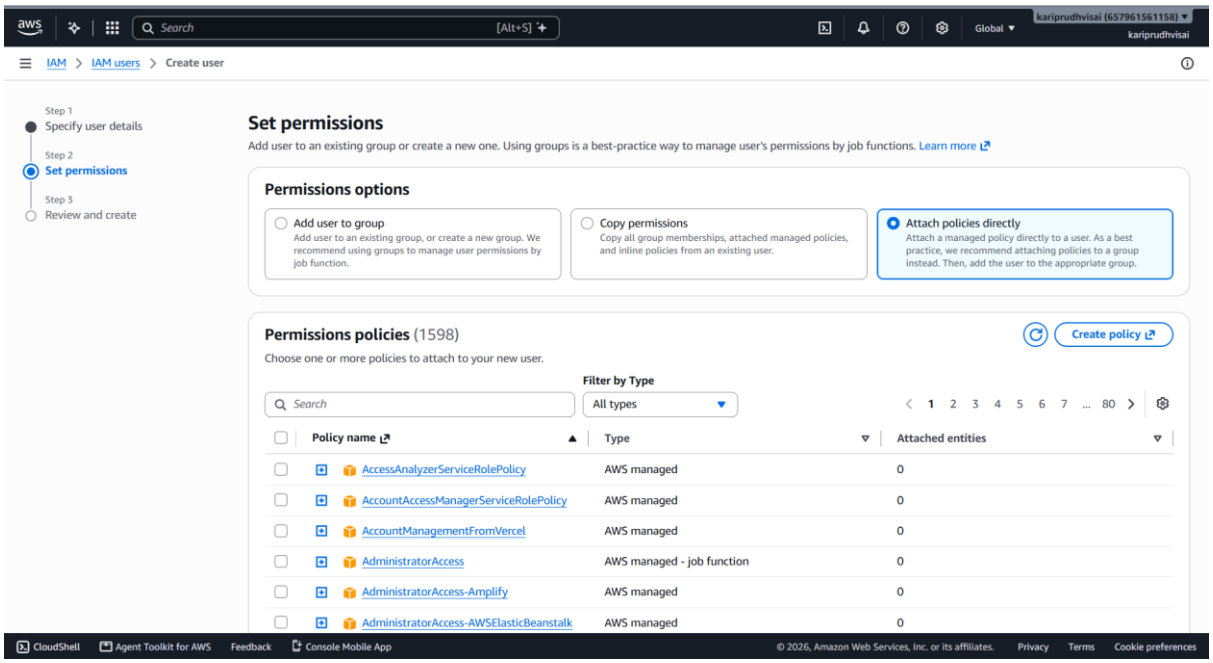
The search bar in the AWS Management Console is used to look up "iam", displaying the IAM service as the top result.



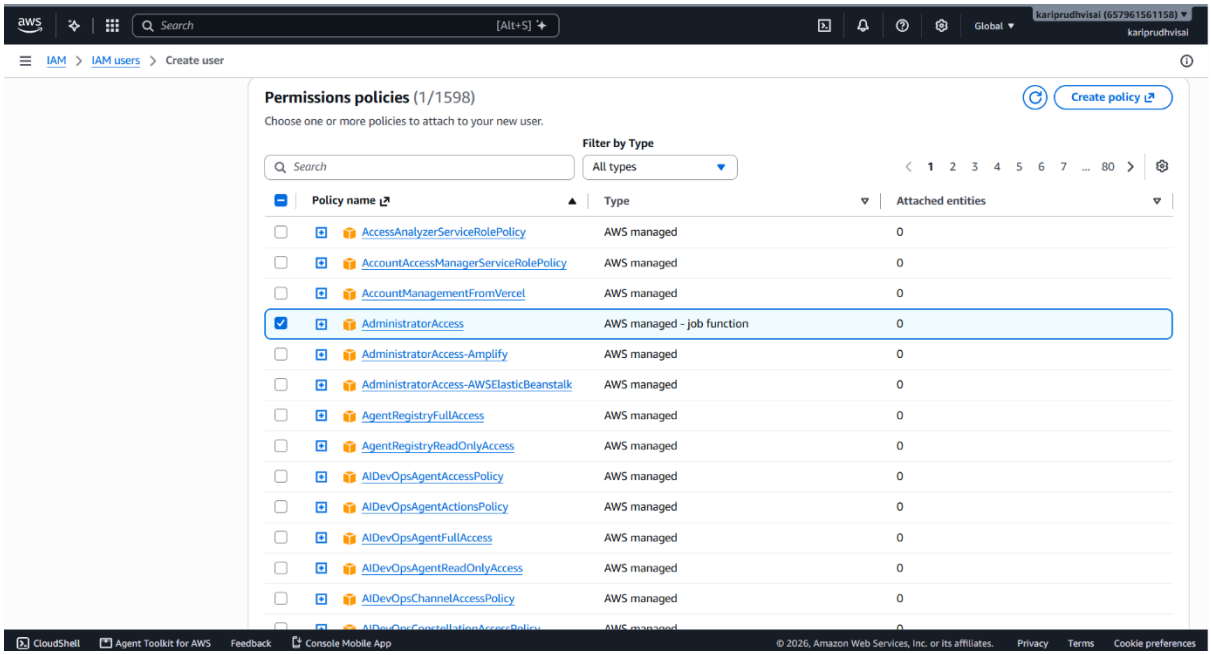
The main "IAM users" dashboard is shown with 0 existing users, displaying a "Create user" button on the top right.



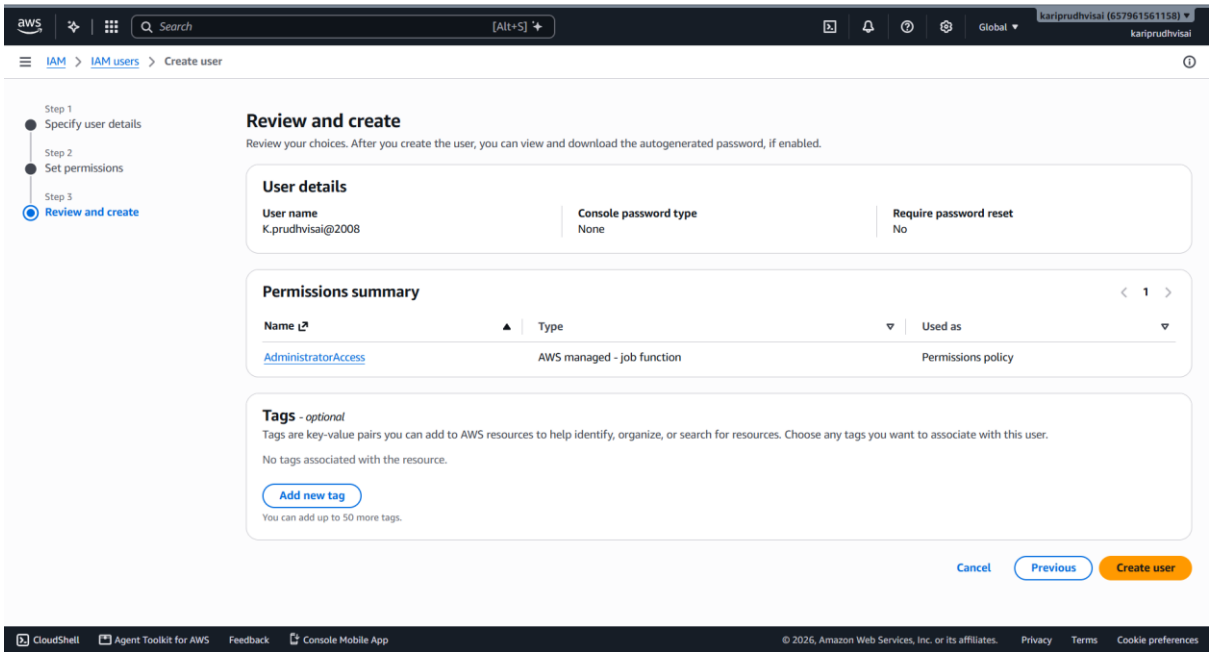
On Step 1 ("Specify user details"), the username "K.prudhvisai@2008" is typed into the User name field.



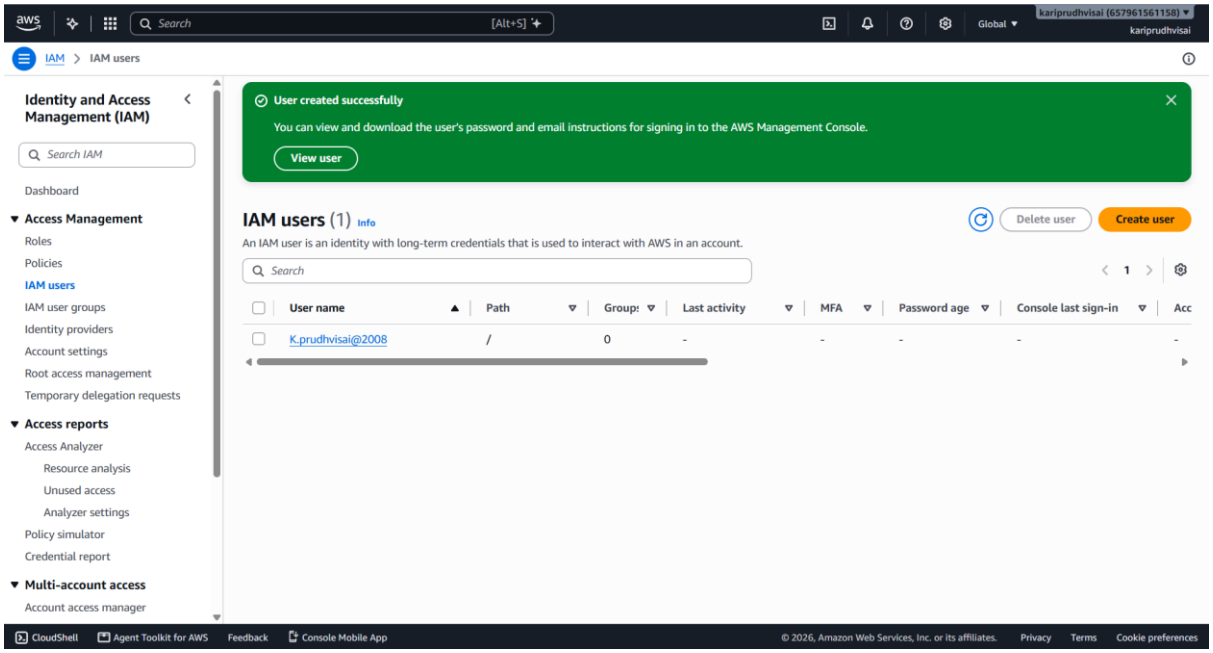
On Step 2 ("Set permissions"), the "Attach policies directly" option is selected from the Permissions options menu.



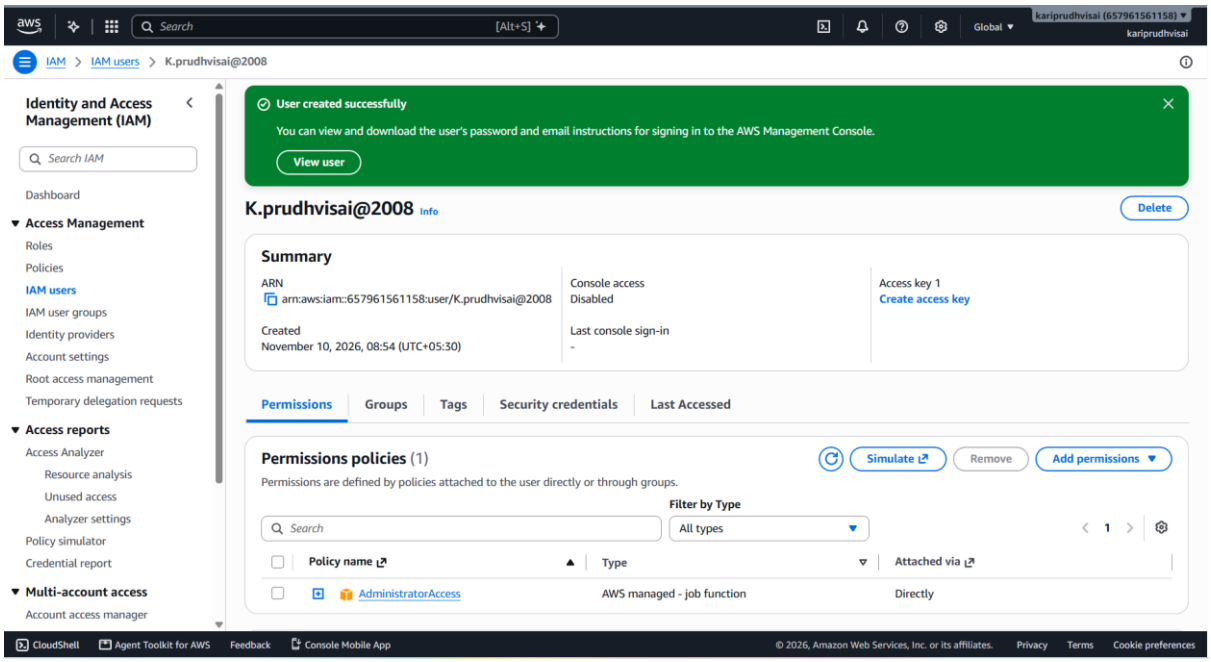
Further down on the Step 2 page, the checkbox next to the "AdministratorAccess" policy is selected to grant full administrative rights.



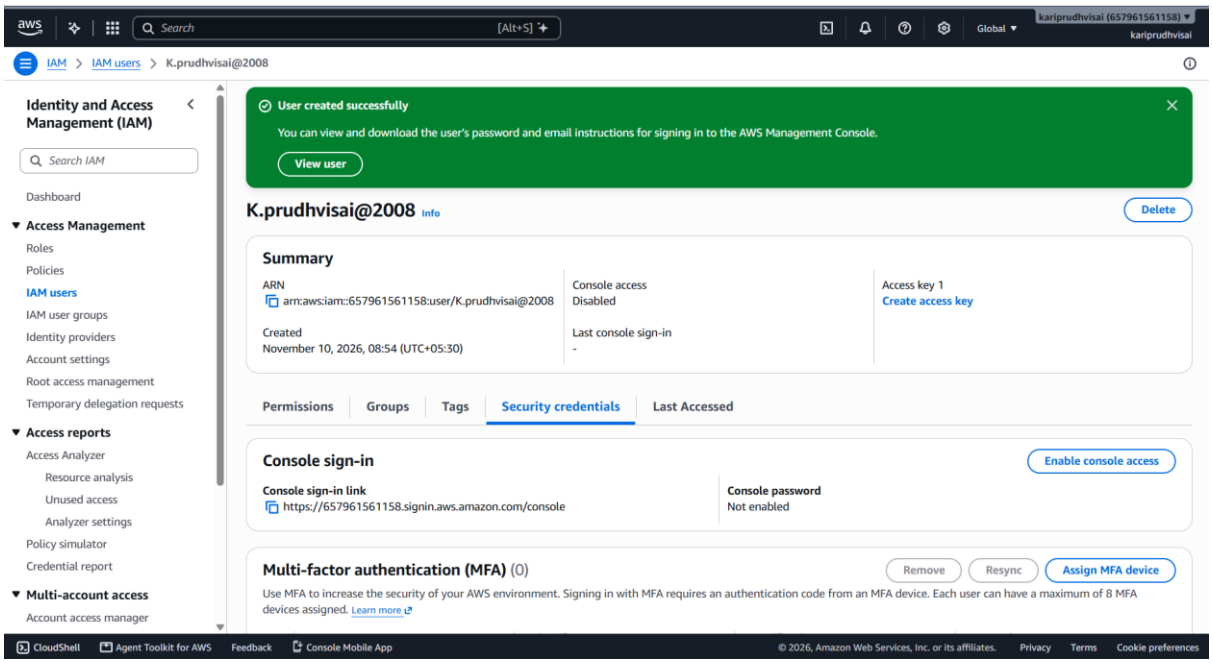
On Step 3 ("Review and create"), a summary page displays the chosen username and the attached AdministratorAccess policy for final verification before clicking "Create user".



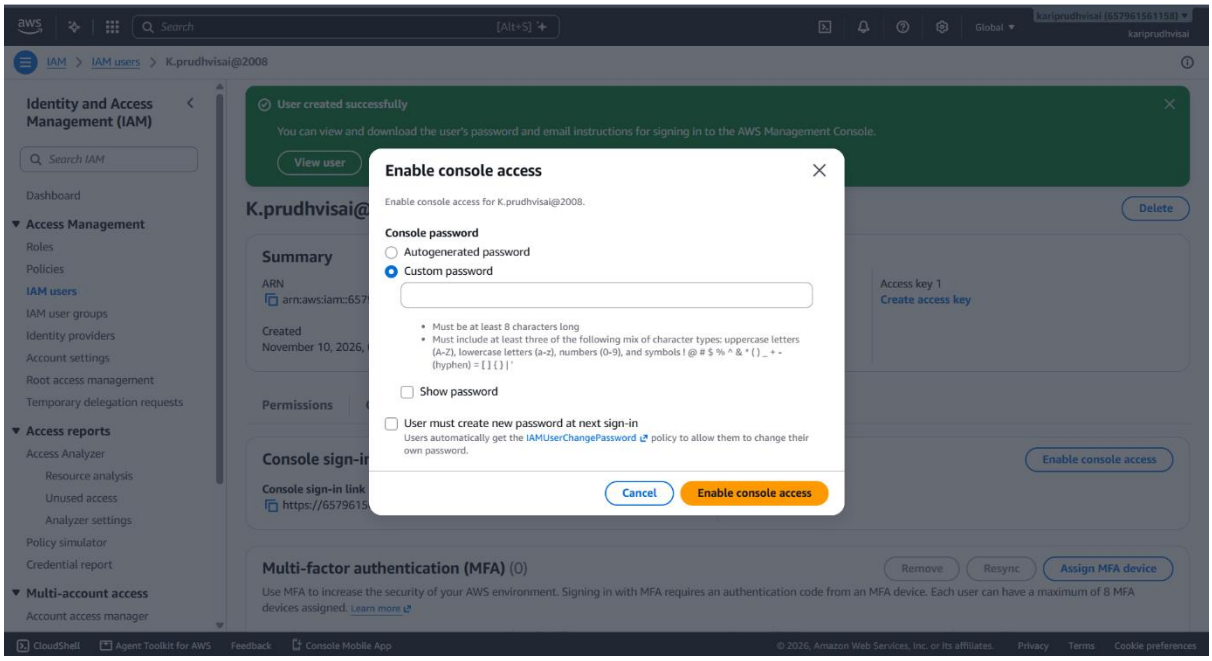
A green banner at the top confirms the user was created successfully, and "K.prudhvisai@2008" is now visible as an entry in the IAM users list.



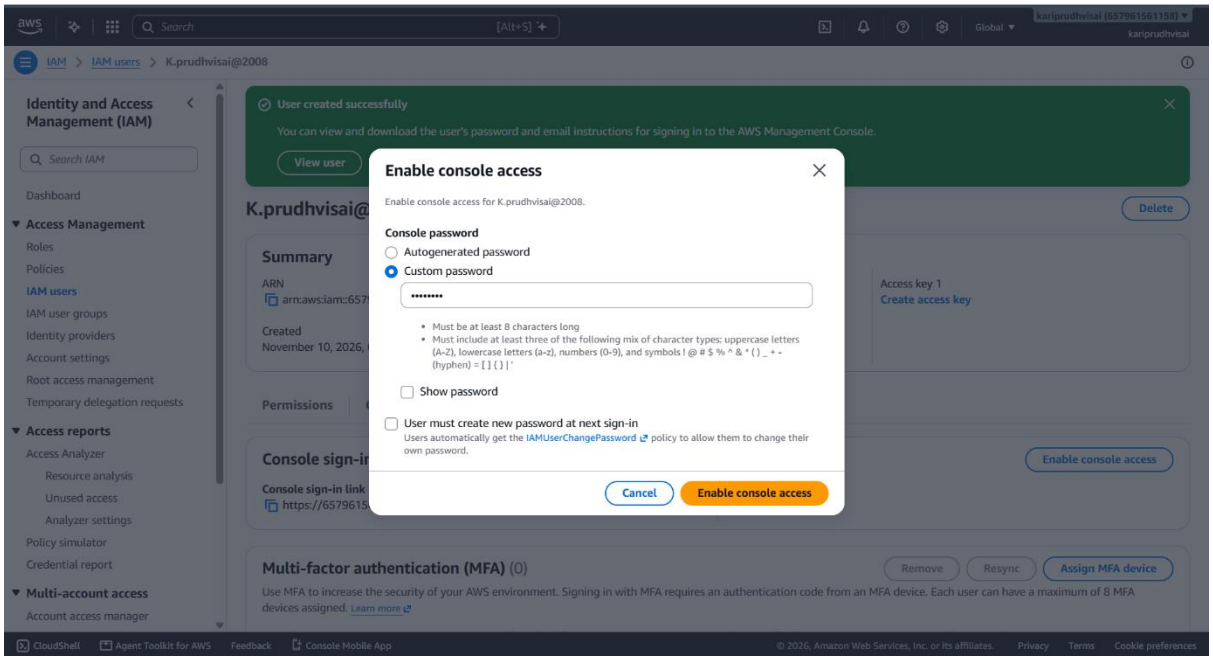
The specific user's detail page shows their Summary information (like their ARN) and confirms the AdministratorAccess policy is attached under the Permissions tab.



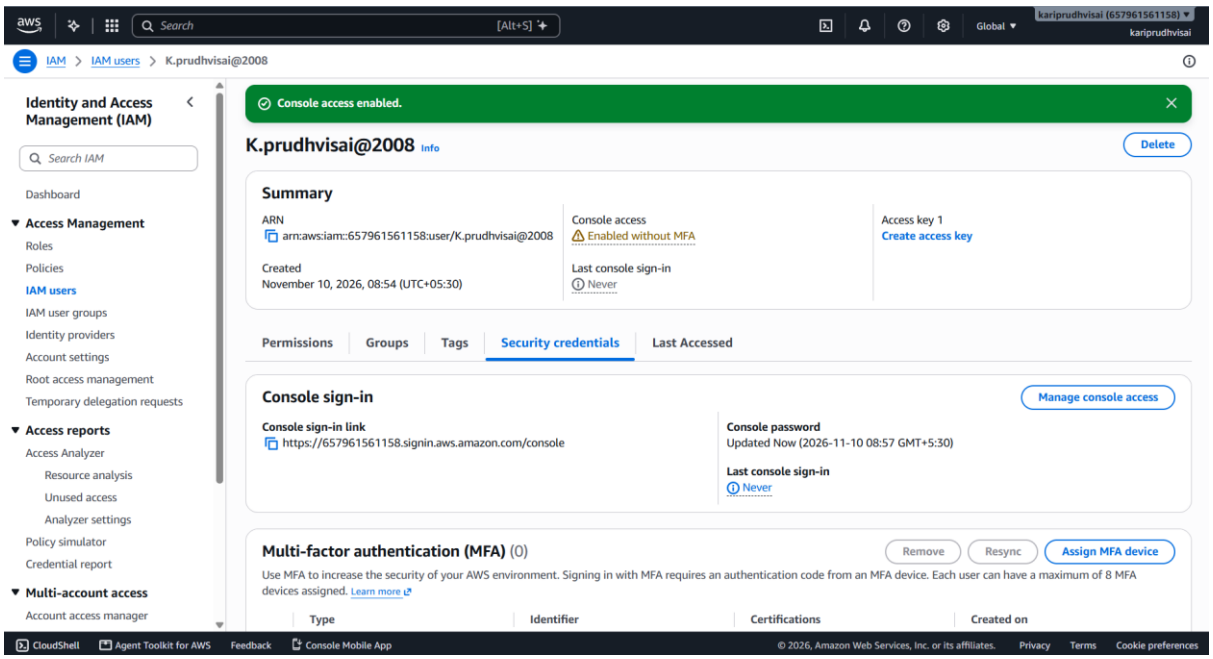
On the same user detail page, the "Security credentials" tab is opened, displaying an "Enable console access" button in the Console sign-in section.



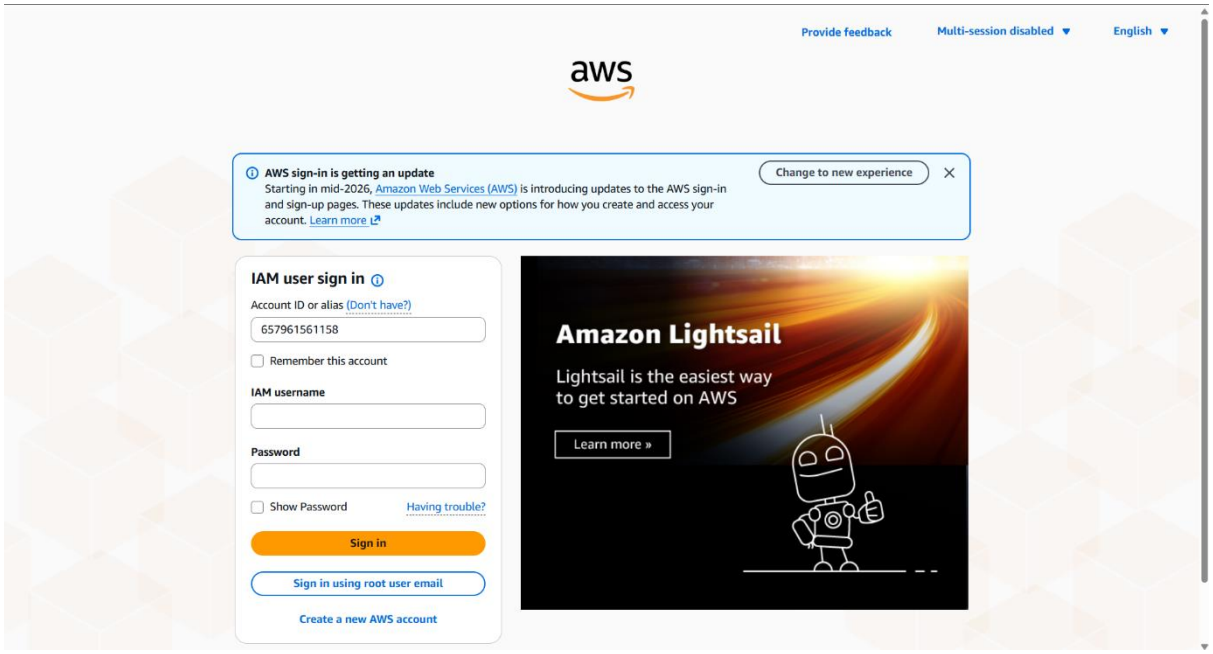
A modal popup titled "Enable console access" appears, where the "Custom password" radio button is selected to manually assign a login password for the user.



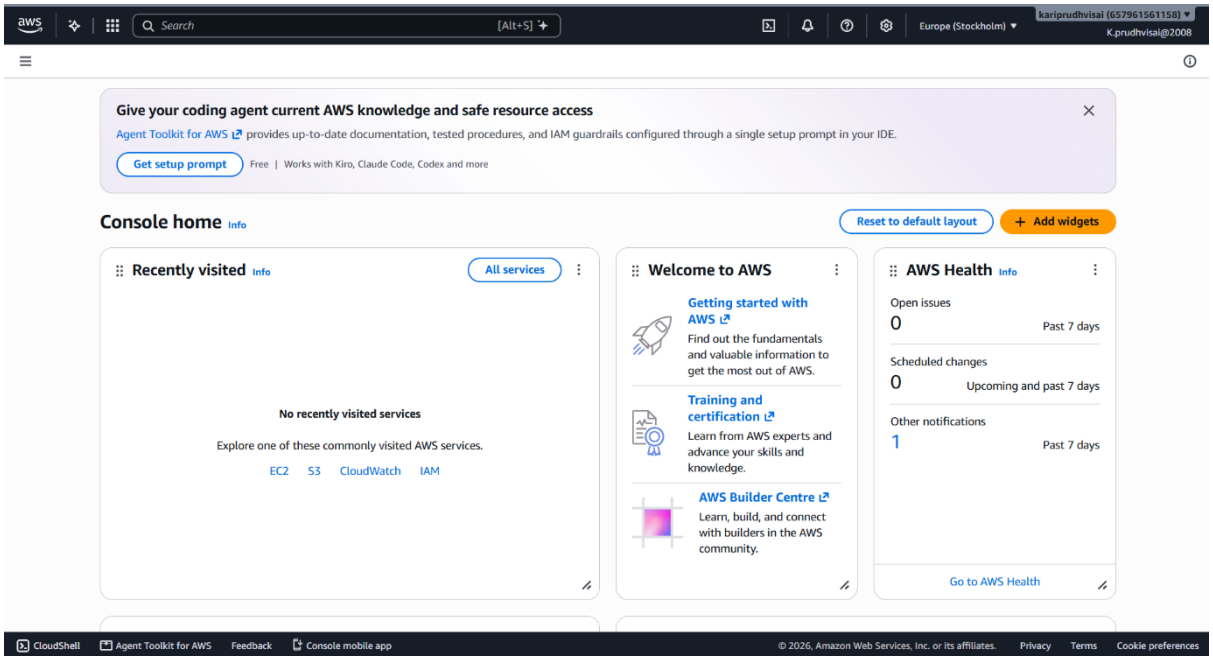
The "Enable console access" modal is active, showing a custom password being entered (hidden as dots). The option requiring the user to create a new password at the next sign-in is left unchecked.



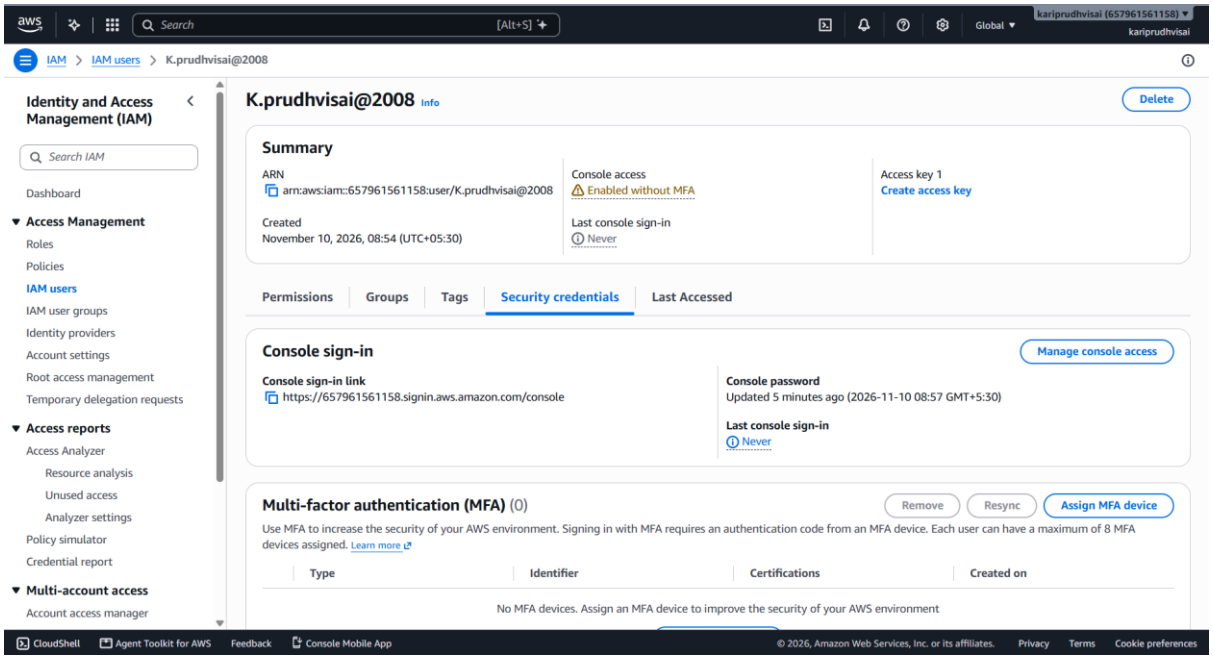
A success window titled "Console password" appears, confirming the password was enabled. It displays the console sign-in URL, the username, a masked view of the password, and a button to download these credentials as a .csv file.



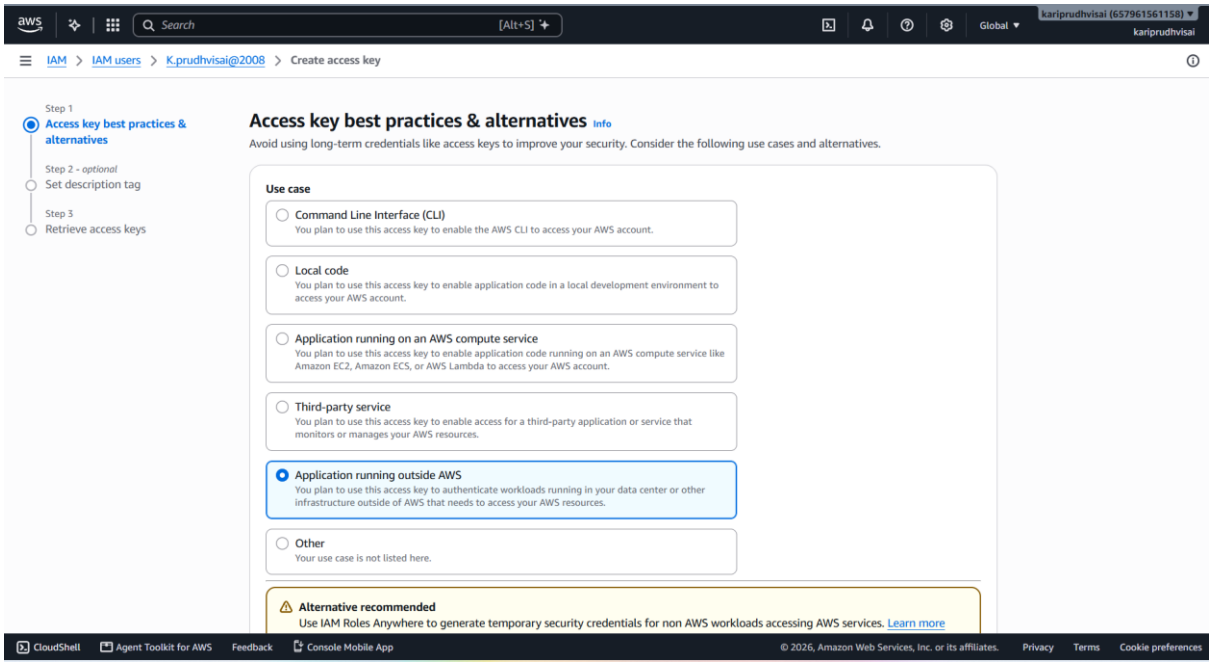
The user's "Security credentials" tab now shows a green banner confirming console access is enabled. The summary details indicate console access is enabled (without MFA) and the password was just updated.



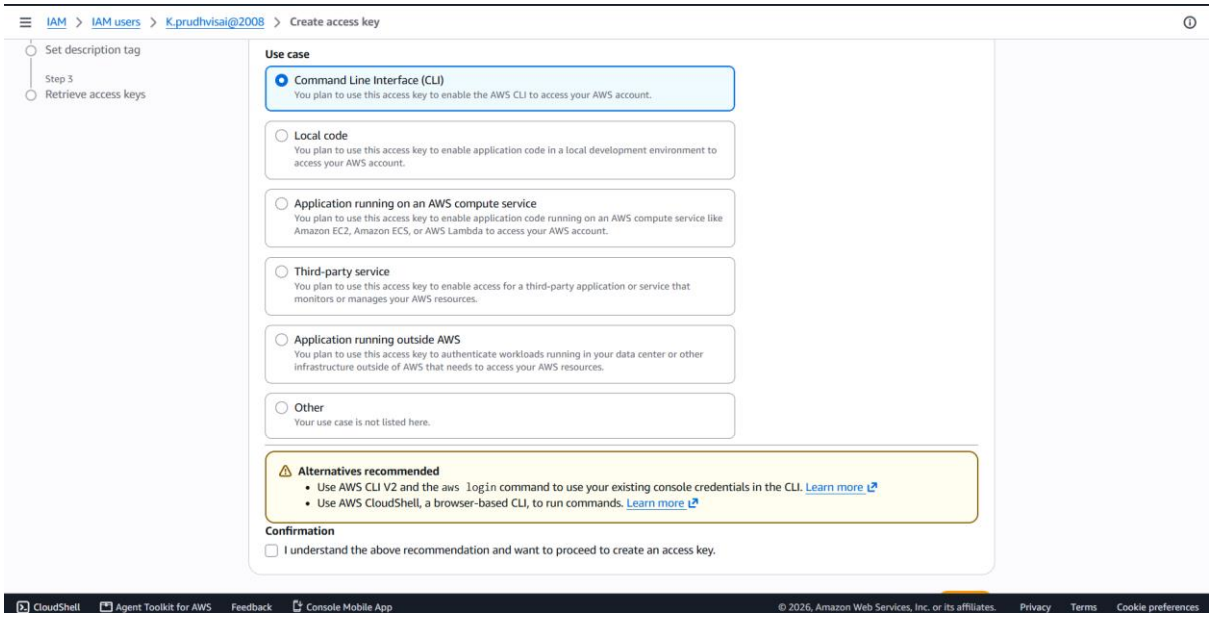
The standard AWS IAM user sign-in screen is displayed. The 12-digit Account ID is pre-filled, waiting for the IAM username and password to be entered to log in.



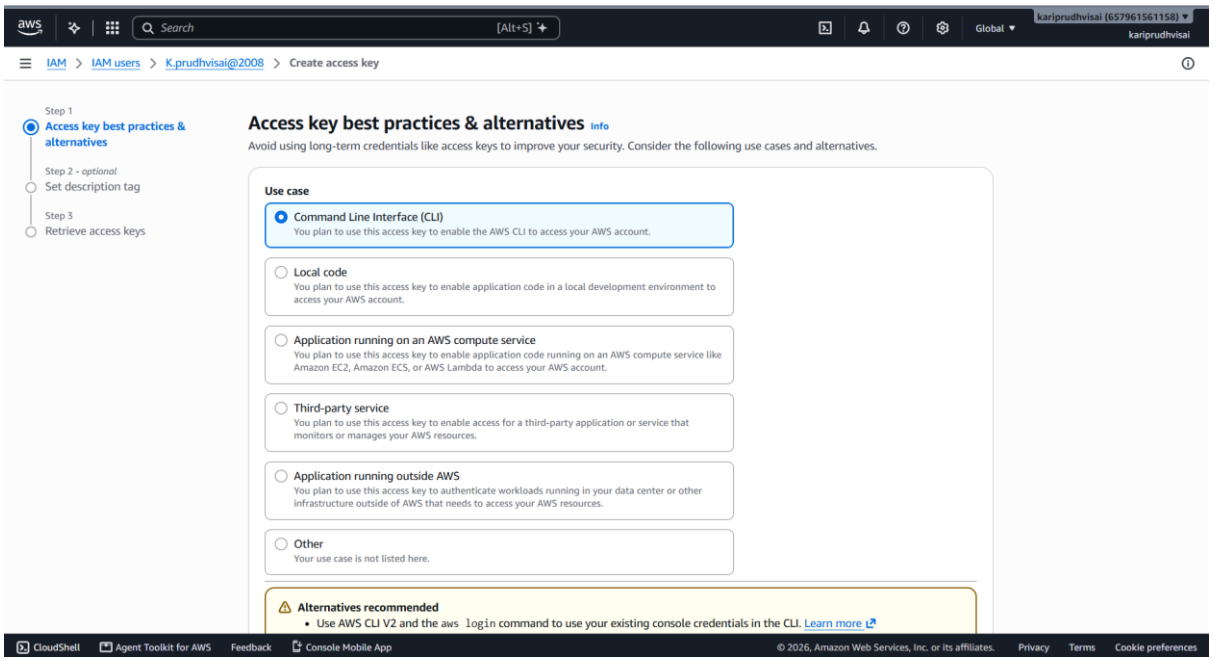
The AWS Console Home dashboard is shown, logged in successfully under the newly created IAM user profile (K.prudhvisai@2008), visible in the top right corner.



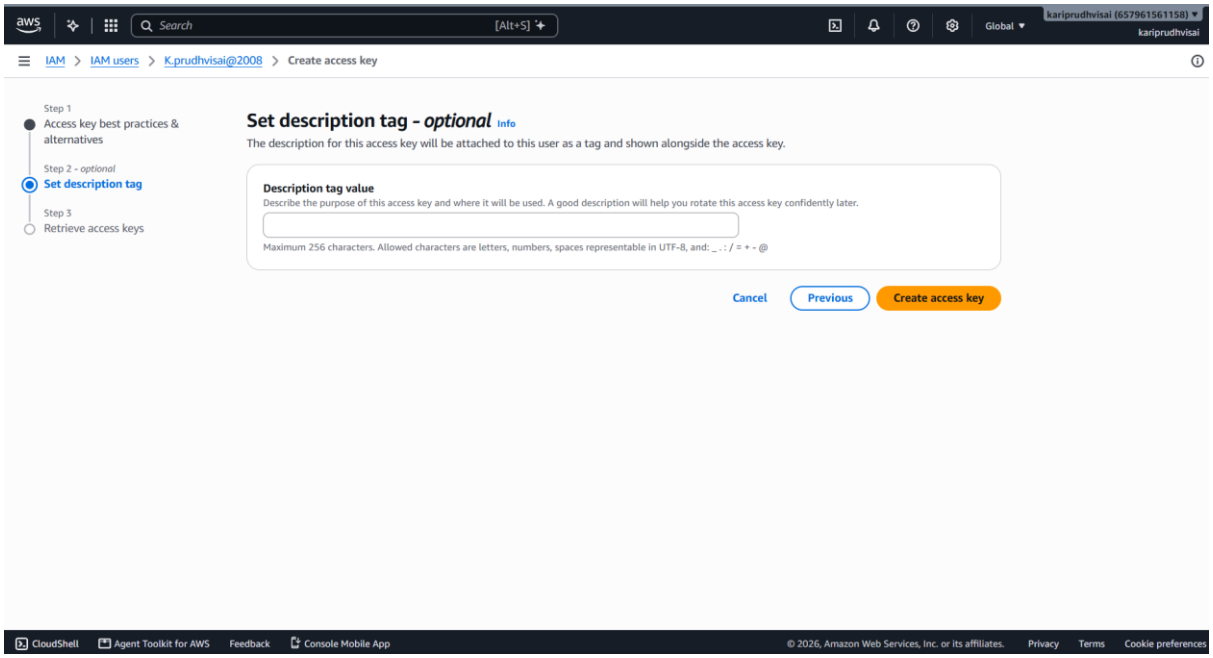
Back on the IAM user's "Security credentials" tab, the interface highlights the "Create access key" button on the right side, preparing to generate programmatic credentials.



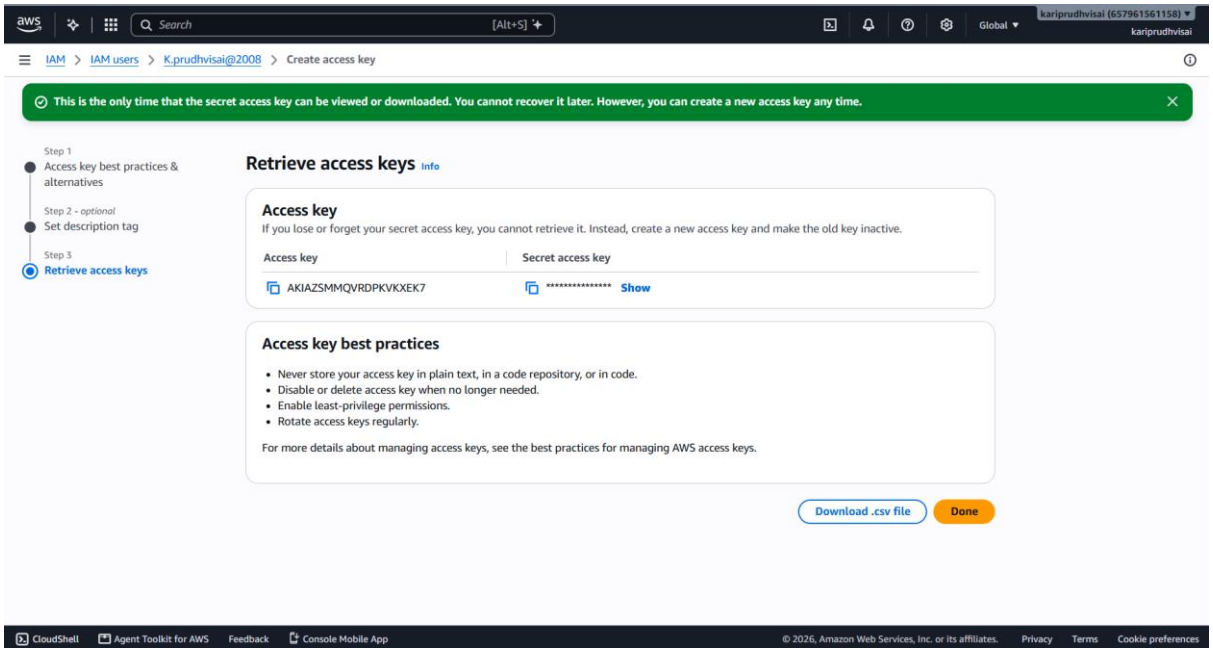
Step 1 of the "Create access key" wizard is shown, where the user must select a use case. The option "Application running outside AWS" is currently selected.



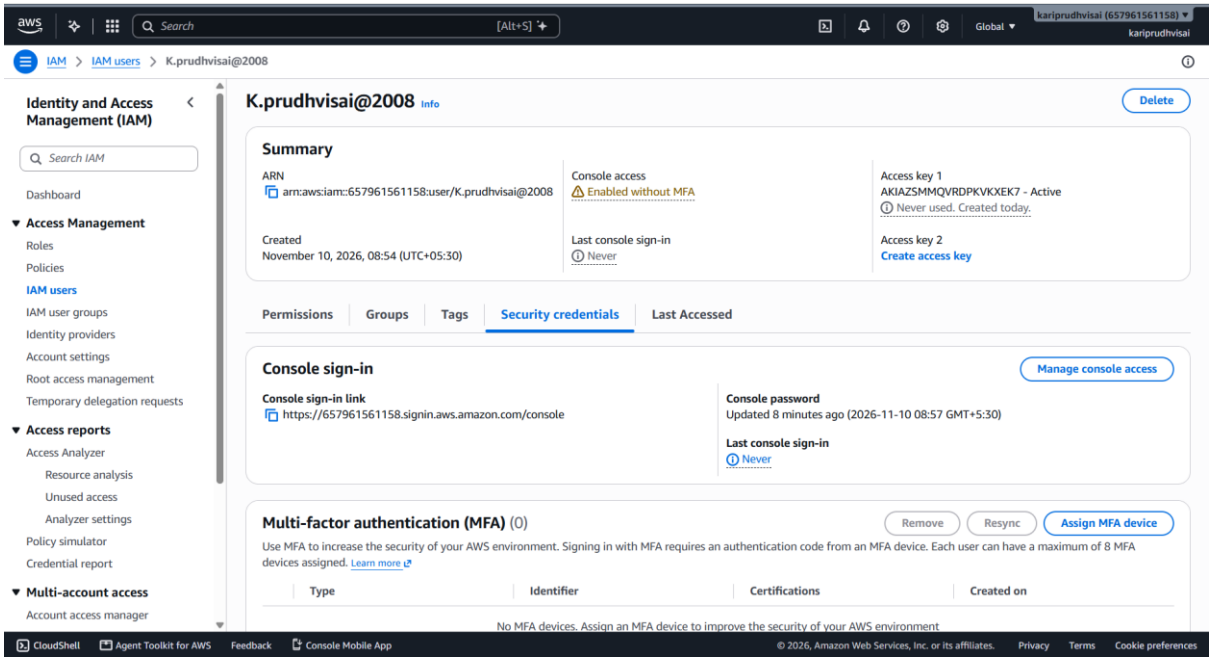
Still on Step 1, the selected use case is changed to "Command Line Interface (CLI)". This triggers an "Alternatives recommended" warning at the bottom, along with a confirmation checkbox that must be ticked to proceed.



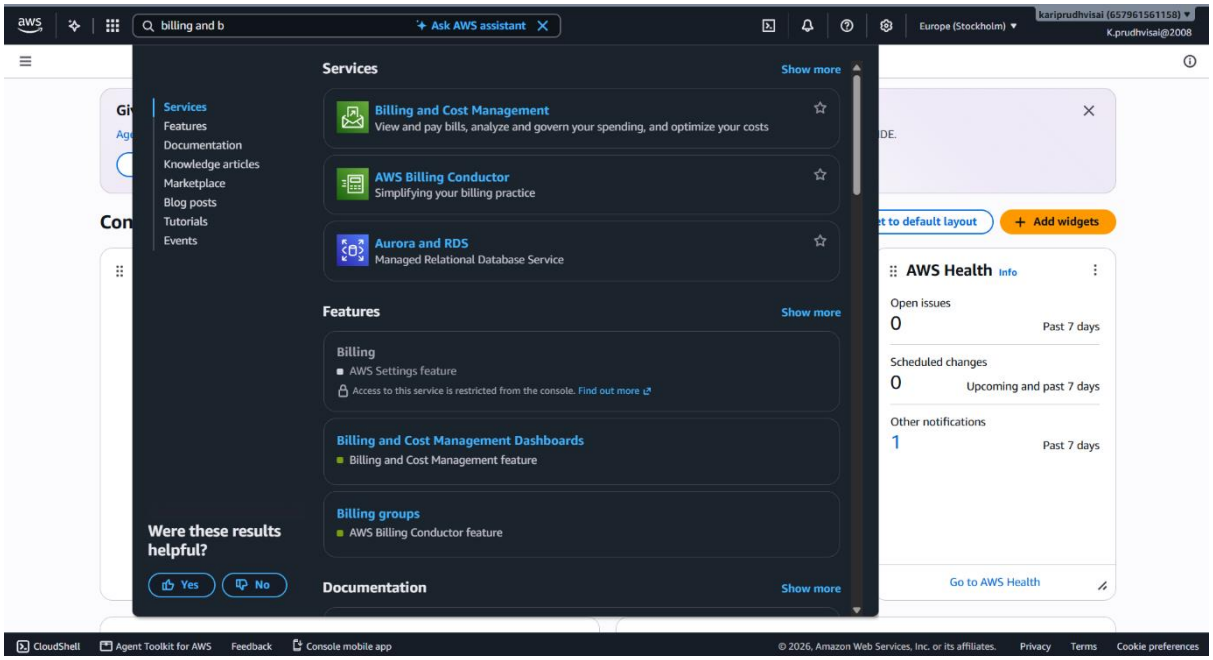
Another view of Step 1, clearly showing the "Command Line Interface (CLI)" option selected as the intended use case for the new access keys.



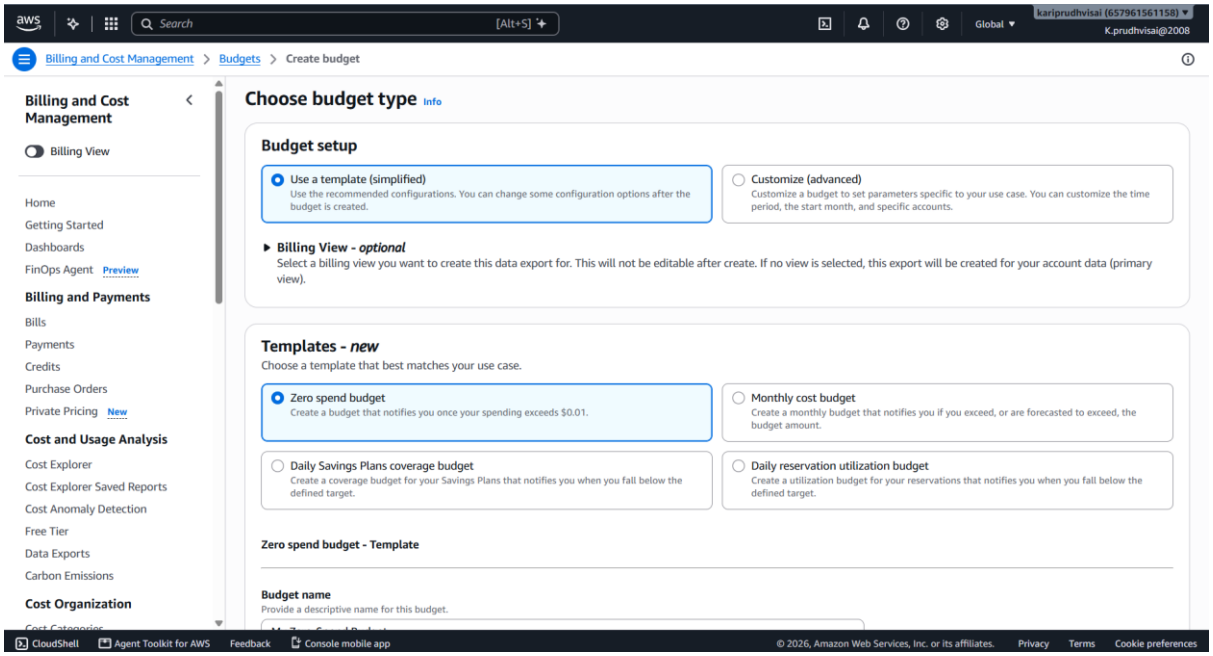
Step 2 of the wizard ("Set description tag") displays an optional text box where a descriptive tag can be added to the access key before clicking the "Create access key" button to finalize the process.



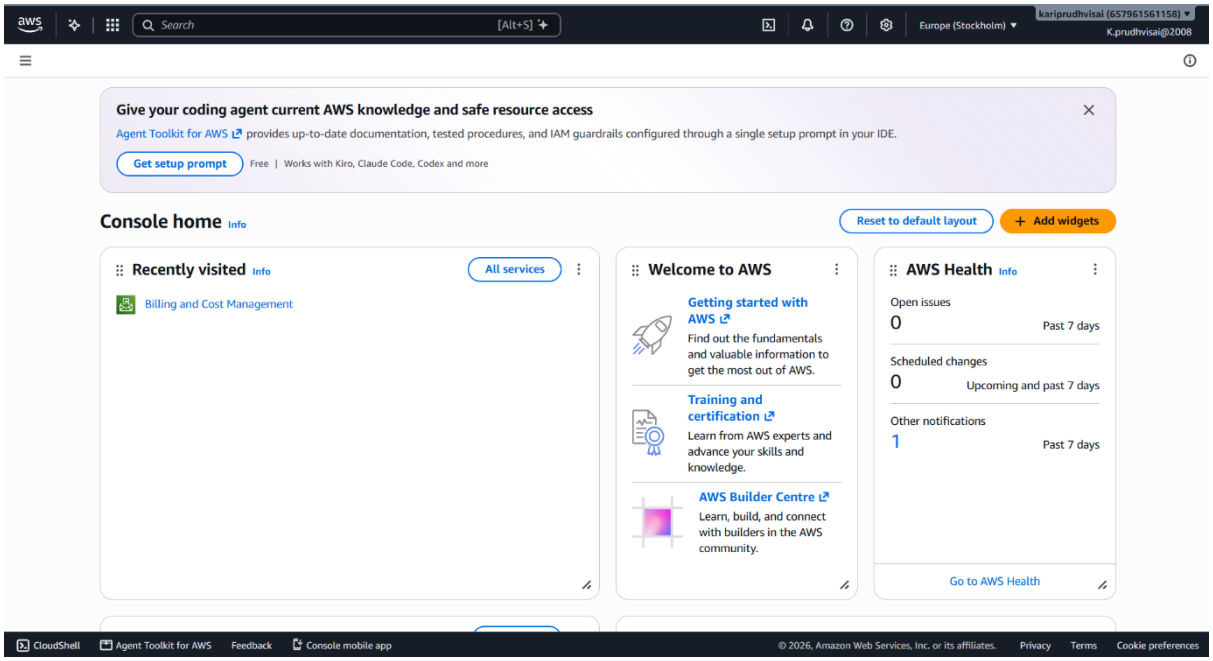
Shows Step 3 of the "Create access key" wizard, titled "Retrieve access keys". It displays the newly generated Access key (AKIAZSMMQVRDPKVKXEK7) and a hidden Secret access key. A green warning banner emphasizes that this is the only time the secret key can be viewed or downloaded before clicking "Done".



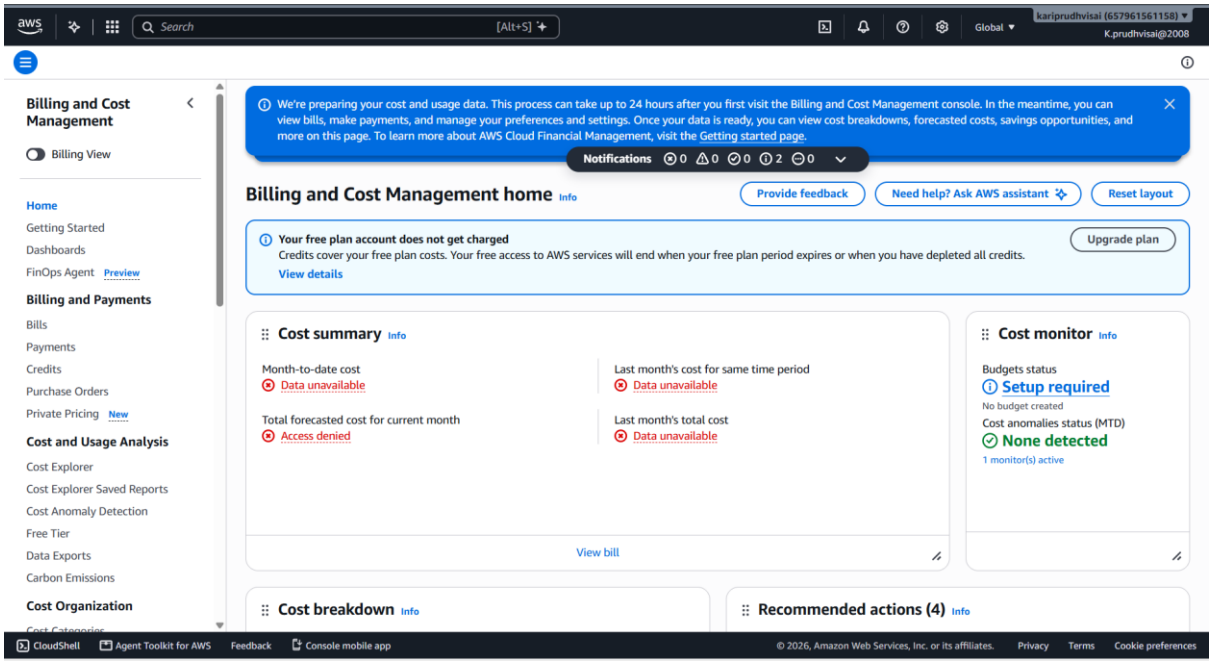
Returns to the IAM user's "Security credentials" tab for the user K.prudhvisai@2008. It confirms the new access key is now listed under the account as "Active" and was "Created today".



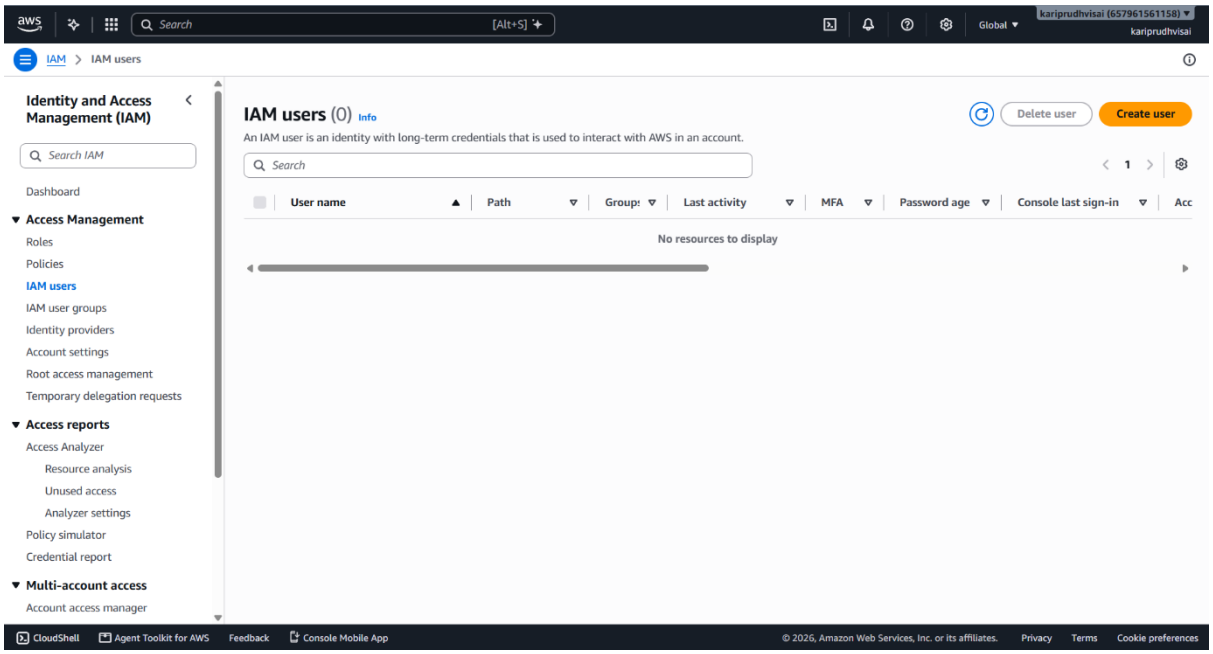
Displays the AWS search bar being used to search for "billing and b". The "Billing and Cost Management" service is highlighted as the top result under Services.



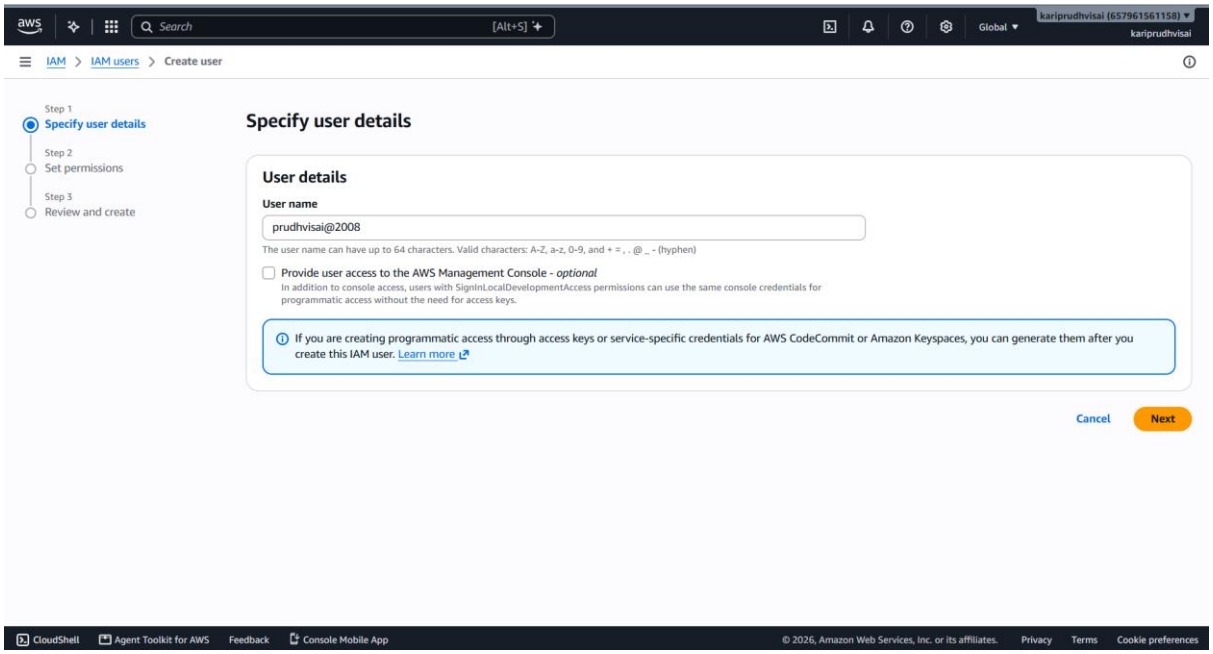
Shows the "Create budget" page within the Billing and Cost Management console. Under "Budget setup", the option "Use a template (simplified)" is selected. Below that, the "Zero spend budget" template is chosen, which is designed to notify the user if spending exceeds $0.01.



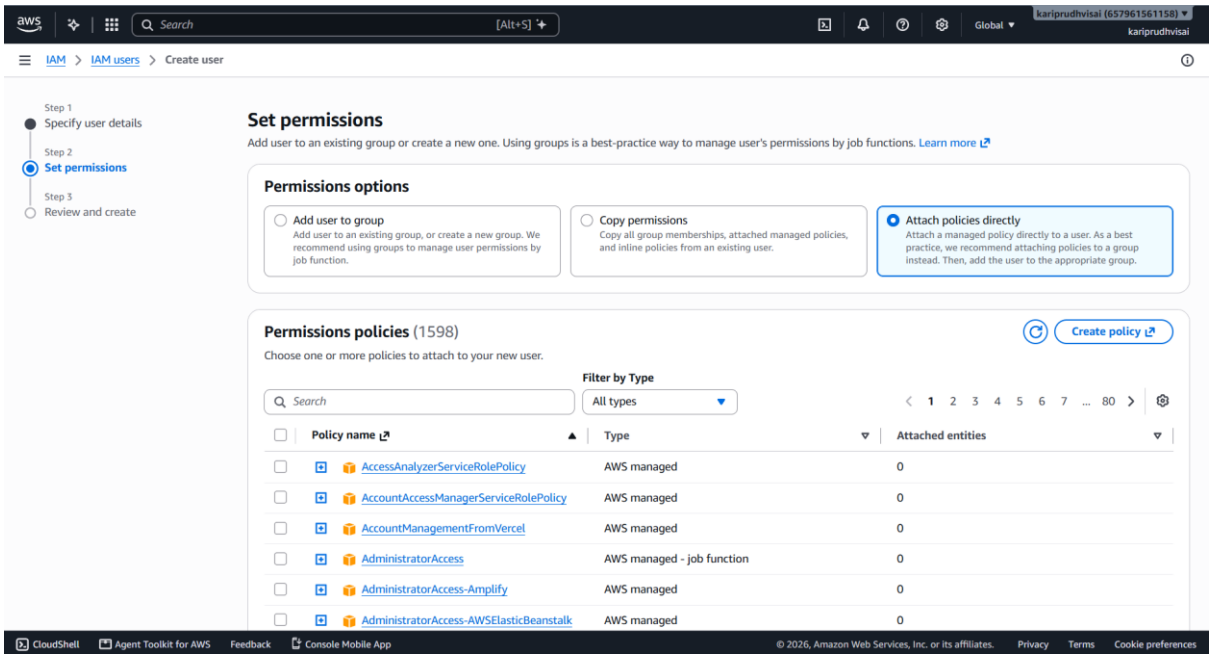
Displays the AWS Console Home dashboard. The "Recently visited" services widget now includes a shortcut to "Billing and Cost Management".



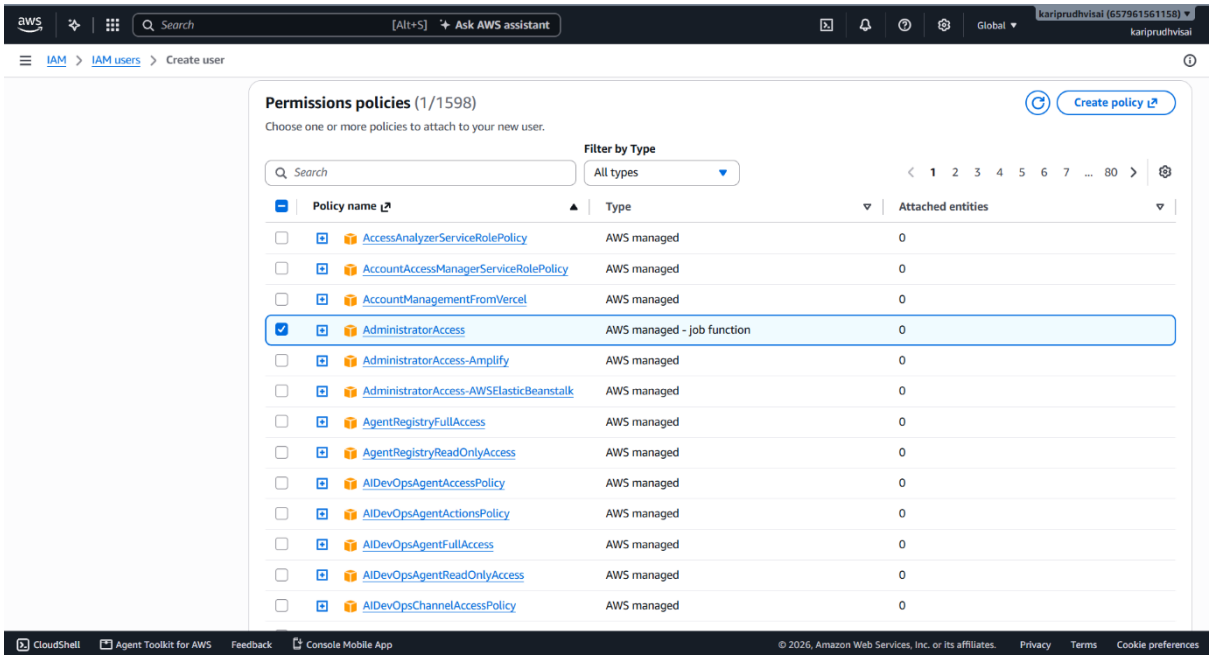
Shows the "Billing and Cost Management home" dashboard. A blue banner states that cost and usage data is being prepared and can take up to 24 hours to become available. The "Cost summary" section displays "Data unavailable" and "Access denied", while the "Budgets status" widget indicates "Setup required" with no budget created.



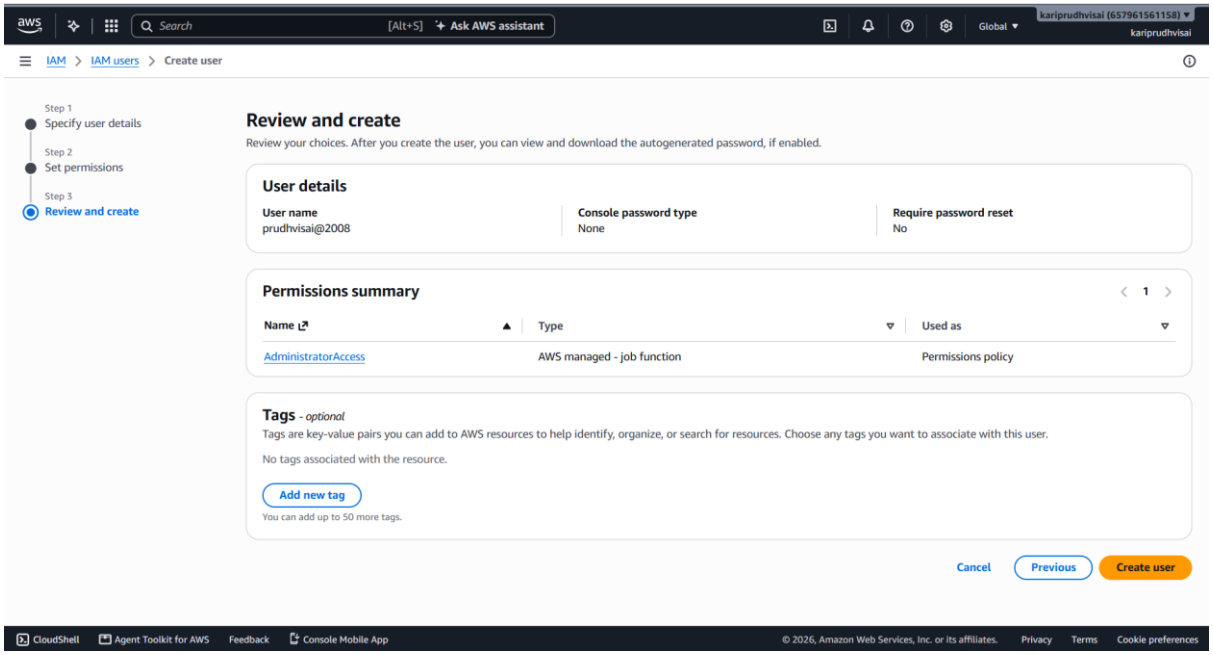
Displays the main "IAM users" dashboard, showing that there are currently 0 IAM users listed in the account.



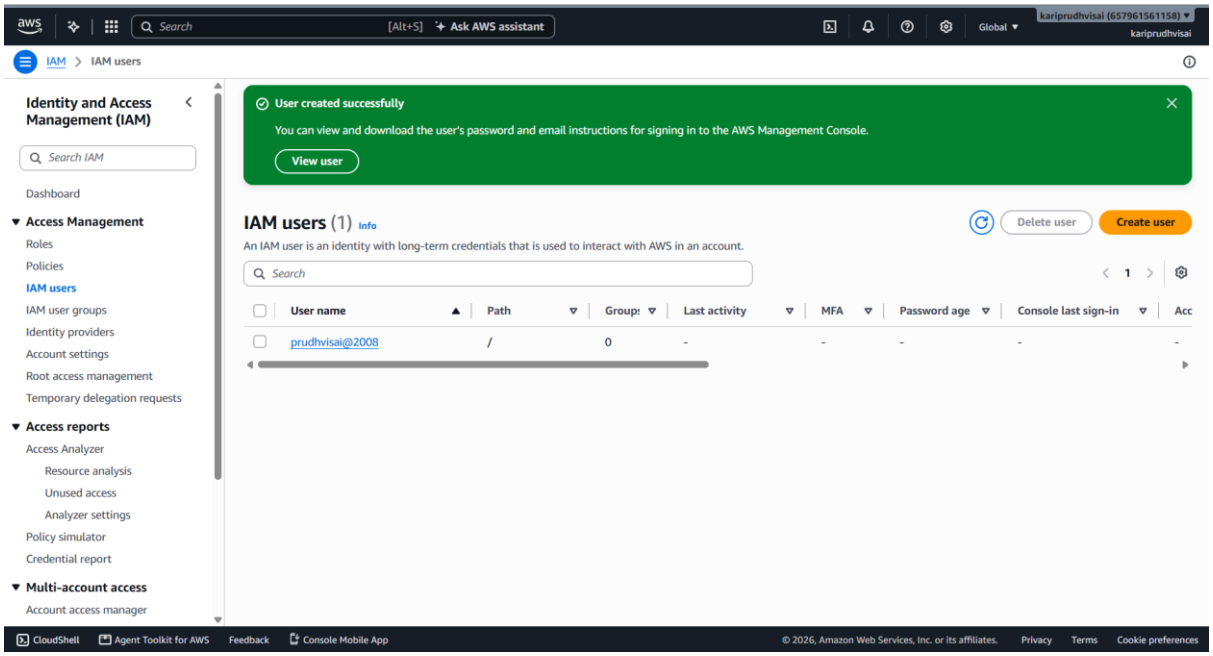
Shows Step 1 ("Specify user details") of a new IAM "Create user" process. The username "prudhvisai@2008" is typed into the User name text field.



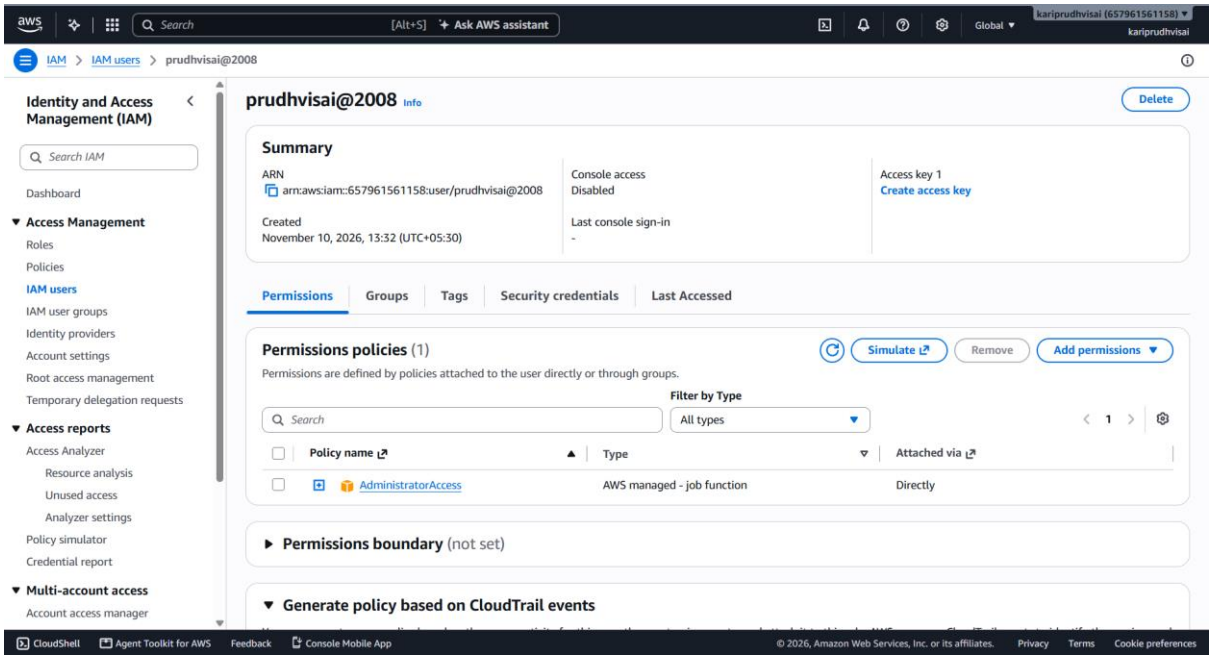
Displays Step 2 ("Set permissions") of creating the new user. The "Attach policies directly" option is selected in the Permissions options box.



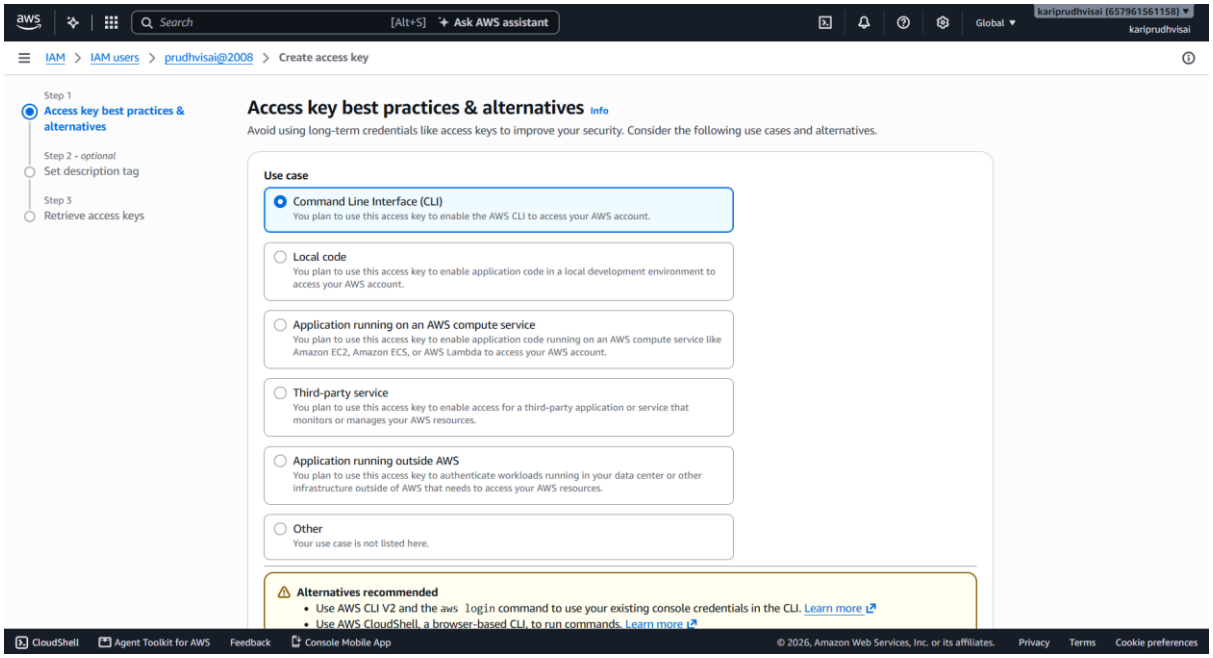
Shows the lower portion of Step 2 ("Set permissions"). The checkbox next to the "AdministratorAccess" policy is selected to grant the new user full administrative privileges.



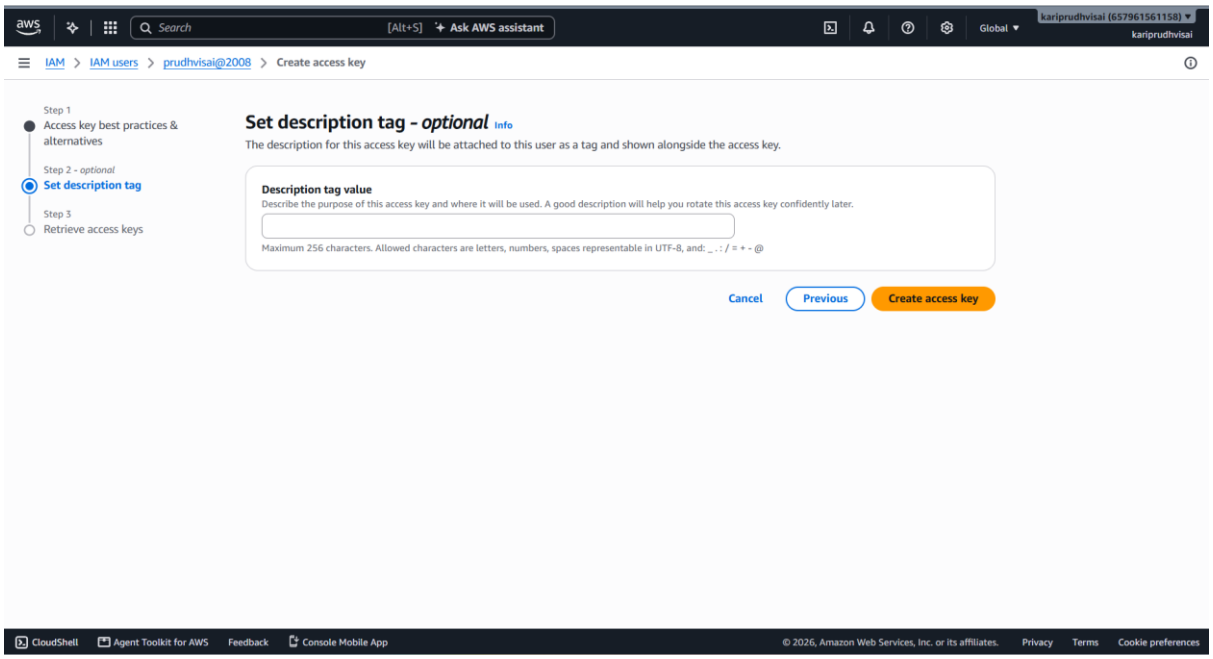
Shows the main "IAM users" dashboard with a green success banner indicating the user was created successfully. The newly created user, prudhvisai@2008, is now visible in the account's user list.



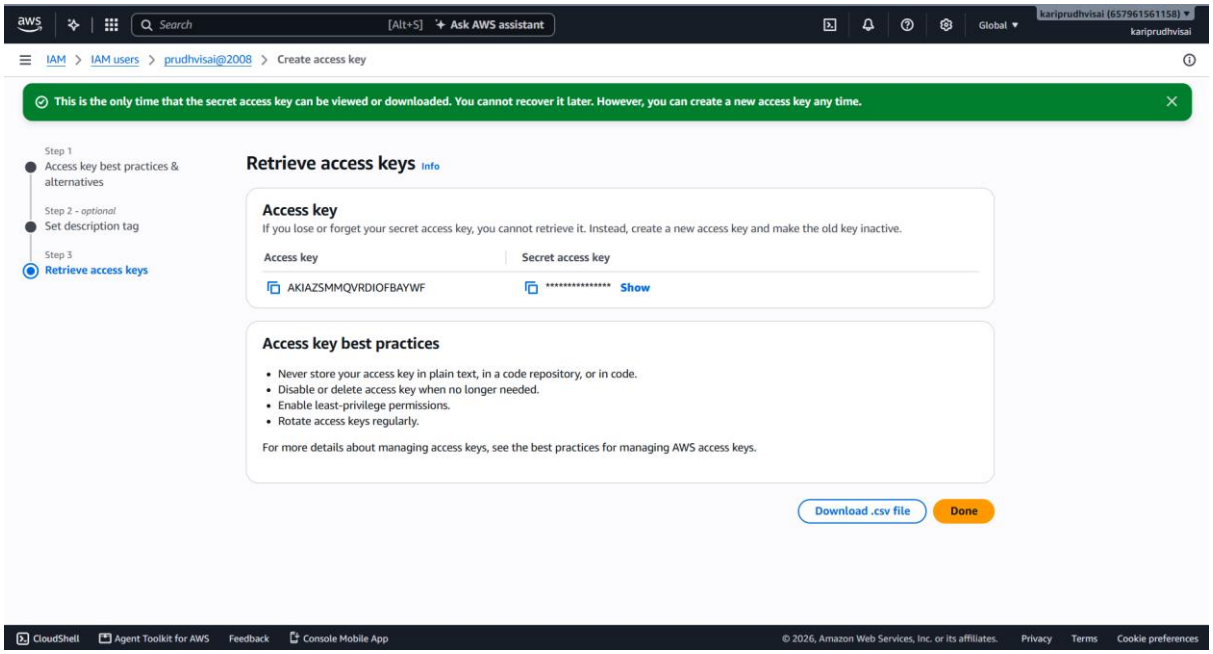
Displays the details page for the user prudhvisai@2008. The "Permissions" tab is active, confirming that the AdministratorAccess policy is attached directly to this account.



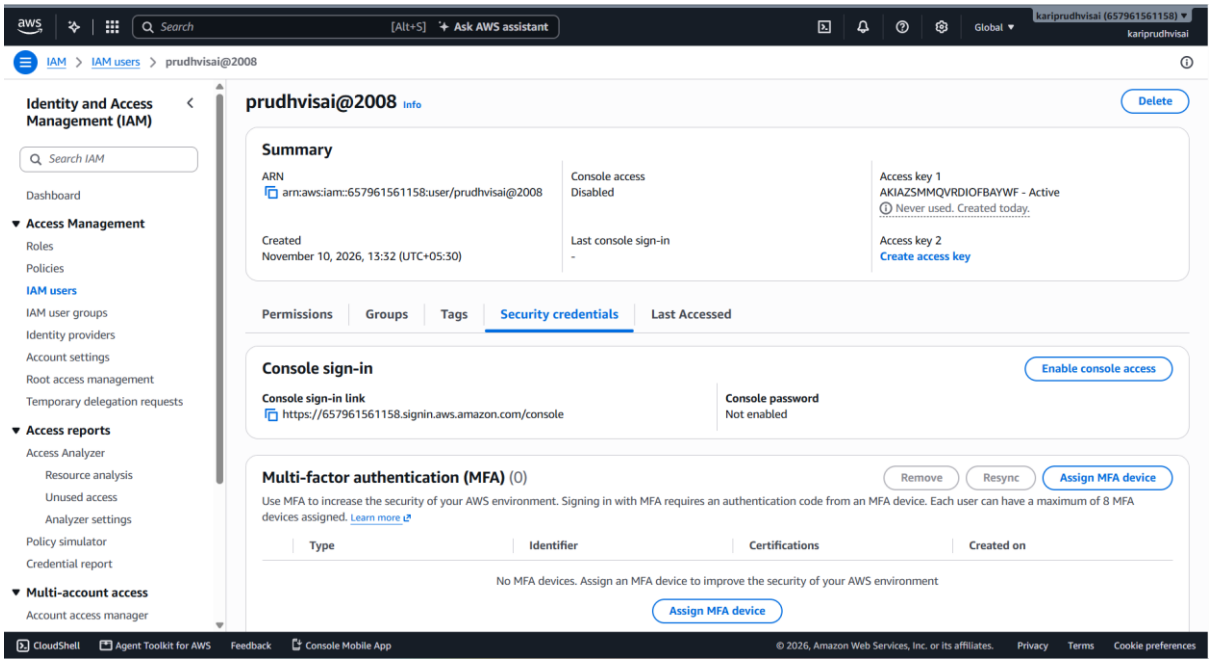
Shows Step 1 ("Access key best practices & alternatives") of the process to create an access key for the user. The "Command Line Interface (CLI)" option is selected as the specific use case for these credentials.



Displays Step 2 ("Set description tag") of the access key creation wizard. The optional description tag text box is left empty before the user finalizes the process by clicking the "Create access key" button.



Shows Step 3 ("Retrieve access keys"), presenting the newly generated Access key (AKIAZSMMQVRDIOFBAYWF) and a masked Secret access key. A prominent green banner warns the user that this is the only opportunity to view or download the secret access key.



Displays the "Security credentials" tab for the user prudhvisai@2008 after the key creation. The summary section confirms that the new Access key 1 (AKIAZSMMQVRDIOFBAYWF) is active and was created today, while noting that console access for this user is currently disabled.